Supplementary material

Participating experts were given as example the FastQC tool annotations, detailed in Figure 4.

Sample tool annotations

```
Tool: FastQC
Topic: DNA-seq, RNA-seq, Sequencing, Sequence analysis,
       Data quality management.
Operation: Quality control, Statistical calculation,
           Sequence composition calculation,
           Diagrams and visualisations.
Data:
    Input => Read sequencing, Raw sequence, Alignments sequence.
    Output => Report sequence.
Format:
    Input => FASTQ/FASTA (text), SAM, BAM.
    Output => HTML.
```

Figure 4: Template given to the experts as a guide to produce annotations on their field-specific life sciences software.

Tool annotation prompt

Context/Role: You are an expert in ontology and life sciences tool annotation. As part of an experiment, you must annotate tools in the field of {{DOMAIN}}.

Objectives: The instructions for annotating the tools are as follows: For each tool, you must annotate according to four themes:

- Topic: the tool's field(s) of application, what it does overall.
- Operation: the actions performed by the tool, how it produces its results.
- Data: types of input and output data.
- Format: input and output file formats.

As an expert, you must choose the most relevant keywords to describe each tool according to the four criteria.

Examples: [...]

Task: Following the format of the annotation examples:

```
Tool: name of the tool
Topic:
Operation:
Data:
    Input =>
    Output =>
Format:
    Input =>
    Output =>
```

Annotation: Annotate the following tools in the field of {{DOMAIN}}:

{{TOOL_NAME}}

Constraints: Use information from the internet to make relevant annotations. Your answer should only contain annotations for this tool, following the examples given. When generating annotations for tools, don't write examples in addition to annotations such as "e.g". Don't write annotations separated by "/", "or", 'and' favour annotations like example: "Protein or gene sequence" → "Protein sequence", "gene sequence"

Figure 5: Prompt given to the LLM agent to perform the life sciences tool annotation, the instruction are mimicking the instructions given to the experts. This prompt includes two placeholders *DOMAIN* and *TOOL_NAME* to specify the target life sciences sub-domain as well as the target life sciences tool. For each tool annotation task, a new DeepSeek "chat" is created. The prompt also includes two annotated tools provided as example which are omitted here for brevity. Examples were truncated in the figure but are present in the GitHub repository : https://github.com/ulysseLeclanche/EDAM_ground_truth/blob/main/Prompt_exemples_two_annotated_tools.txt